

MOG2 + Lucas-Kanade
($T=14.2$ ms, $P=97.0\%$)

MOG2 + SORT
($T=16.55$ ms, $P=96.1\%$)

Capture Probability P_{capture} (%)

MOG2 + IMM (2-Model)
($T=22.1$ ms, $P=92.7\%$)

STMSS (Proposed)
($T=18.82$ ms, $P=95.2\%$)

YOLOv8n + ByteTrack
($T=47.29$ ms, $P=53.4\%$)

— Deterministic P_{capture} Bound
--- 95% Biosecurity Threshold
..... $T_{\text{allow}} = 75.0$ ms Limit

Isolated Pipeline Latency T_{comp} (ms)

